

Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

ANSWER KEY



Domain and range of arcsin

- 1 State the domain and range of $y = \arcsin x$.
Domain: $-1 \leq x \leq 1$. Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.
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Domain and range of arccos

- 2 State the domain and range of $y = \arccos x$.
Domain: $-1 \leq x \leq 1$. Range: $0 \leq y \leq \pi$.
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Domain and range of arctan

- 3 State the domain and range of $y = \arctan x$.
Domain: all real numbers. Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$.
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Evaluating inverse trigonometric functions exactly

- 4 Find the exact value of $\arcsin(\sqrt{3}/2)$, in radians.
 $\arcsin(\sqrt{3}/2) = \pi/3$, since $\sin(\pi/3) = \sqrt{3}/2$ and $\pi/3$ lies in the range $[-\pi/2, \pi/2]$.
- 5 Find the exact value of $\arccos(-1/2)$, in radians.
 $\arccos(-1/2) = 2\pi/3$, since $\cos(2\pi/3) = -1/2$ and $2\pi/3$ lies in the range $[0, \pi]$.
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Differentiating arcsin using the chain rule

- 6 Differentiate $y = \arcsin(3x)$ with respect to x .
Using $\frac{d}{dx} \arcsin(u) = \frac{u'}{\sqrt{1-u^2}}$ with $u = 3x$, $u' = 3$: $\frac{dy}{dx} = \frac{3}{\sqrt{1-9x^2}}$.
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Differentiating arctan using the chain rule

- 7 Differentiate $y = \arctan(x^2)$ with respect to x .
Using $\frac{d}{dx} \arctan(u) = \frac{u'}{1+u^2}$ with $u = x^2$, $u' = 2x$: $\frac{dy}{dx} = \frac{2x}{1+x^4}$.
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Differentiating a product involving arccos

- 8 Differentiate $y = x \arccos(x)$ with respect to x .

By the product rule, $\frac{dy}{dx} = 1 \cdot \arccos(x) + x \cdot \frac{d}{dx} \arccos(x)$. Since $\frac{d}{dx} \arccos(x) = -1/\sqrt{1-x^2}$, the second term is $-x/\sqrt{1-x^2}$. So $\frac{dy}{dx} = \arccos(x) - \frac{x}{\sqrt{1-x^2}}$.

Finding a gradient at a point

- 9 Find the gradient of the curve $y = \arcsin(x/2)$ at $x = 1$.

Simplifying first: $\frac{dy}{dx} = \frac{1/2}{\sqrt{1-x^2/4}} = \frac{1}{\sqrt{4-x^2}}$ (multiplying top and bottom by 2). At $x = 1$, this is $\frac{1}{\sqrt{3}}$, i.e. $\sqrt{3}/3$.

Deriving the derivative of arctan from first principles

- 10 If $y = \arctan x$, then $x = \tan y$. Differentiate implicitly with respect to x to show that $\frac{dy}{dx} = \frac{1}{1+x^2}$.
Let $y' = \frac{dy}{dx}$. Differentiating $x = \tan y$ implicitly with respect to x gives $1 = \sec^2 y \cdot y'$. Since $\sec^2 y = 1 + \tan^2 y = 1 + x^2$ (using $\tan y = x$), this becomes $1 = (1+x^2)y'$, so $\frac{dy}{dx} = \frac{1}{1+x^2}$, as required.
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Identifying the graph of arctan

- 11 Which of the following best describes the graph of $y = \arctan x$?

- a. An odd function with horizontal asymptotes $y = \pm \frac{\pi}{2}$ ✓
- b. An even function with a vertical asymptote at $x = 0$
- c. A periodic function with period 2π
- d. A function with domain restricted to $[-1, 1]$

$\arctan x$ is odd ($\arctan(-x) = -\arctan x$) and, as $x \rightarrow \pm \infty$, $\arctan x \rightarrow \pm \frac{\pi}{2}$, so its graph has horizontal (not vertical) asymptotes at $y = \pm \frac{\pi}{2}$ and its domain is all real numbers, not $[-1, 1]$.

Tangent line to an inverse trigonometric curve

- 12 Find the equation of the tangent line to $y = \arcsin x$ at $x = 0$.

At $x = 0$: $y = \arcsin(0) = 0$. The derivative is $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$, which at $x = 0$ gives gradient 1. The tangent line is $y - 0 = 1(x - 0)$, i.e. $y = x$.

Evaluating arctan exactly

- 13 Find the exact value of $\arctan(1)$, in radians.

$\arctan(1) = \pi/4$, since $\tan(\pi/4) = 1$ and $\pi/4$ lies in the range $(-\pi/2, \pi/2)$.

Differentiating arctan of a linear expression

- 14 Differentiate $y = \arctan(5x)$ with respect to x .

Using $d/dx \arctan(u) = u'/(1 + u^2)$ with $u = 5x$, $u' = 5$: $dy/dx = 5/(1 + 25x^2)$.

Differentiating arcsin of a cubic expression

- 15 Differentiate $y = \arcsin(x^3)$ with respect to x .

Using $d/dx \arcsin(u) = u'/\sqrt{1 - u^2}$ with $u = x^3$, $u' = 3x^2$: $dy/dx = 3x^2/\sqrt{1 - x^6}$.

Evaluating a derivative at a specific point

- 16 Find the gradient of $y = \arccos(2x)$ at $x = 0$.

Using $d/dx \arccos(u) = -u'/\sqrt{1 - u^2}$ with $u = 2x$, $u' = 2$: $dy/dx = -2/\sqrt{1 - 4x^2}$. At $x = 0$, this is $-2/\sqrt{1} = -2$.

Tangent line at a non-zero point

- 17 Find the equation of the tangent line to $y = \arctan x$ at $x = 1$.

At $x = 1$: $y = \arctan(1) = \pi/4$. The derivative is $dy/dx = 1/(1 + x^2)$, which at $x = 1$ gives gradient $1/2$. The tangent line is $y - \pi/4 = (1/2)(x - 1)$.

Differentiating arctan of a reciprocal expression

- 18 Differentiate $y = \arctan(1/x)$ with respect to x , and evaluate the derivative at $x = 1$.

Let $u = 1/x$, so $u' = -1/x^2$. Then $dy/dx = u'/(1 + u^2) = (-1/x^2)/(1 + 1/x^2)$. Multiplying top and bottom by x^2 gives $dy/dx = -1/(x^2 + 1)$. At $x = 1$: $dy/dx = -1/2$.

Solving an equation involving arcsin

- 19 Solve $\arcsin(x) = \pi/6$ for x .

$x = \sin(\pi/6) = 1/2$.

Solving an equation involving arccos

- 20 Solve $\arccos(x) = 2\pi/3$ for x .

$$x = \cos(2\pi/3) = -1/2.$$

Explaining why an inverse trig expression is undefined

- 21 Explain why $\arcsin(2)$ is undefined.

The domain of $\arcsin$ is $[-1, 1]$, since sine itself never takes a value outside $[-1, 1]$. Because 2 lies outside this domain, there is no angle whose sine equals 2, so $\arcsin(2)$ is undefined.

Proving the arcsin-arccos complementary identity

- 22 Prove that $\arcsin(x) + \arccos(x) = \pi/2$ for every x in $[-1, 1]$.

Let $\theta = \arcsin(x)$, so $\sin \theta = x$ with θ in $[-\pi/2, \pi/2]$. By the co-function identity, $\cos(\pi/2 - \theta) = \sin \theta = x$. Since $\theta \in [-\pi/2, \pi/2]$, the angle $\pi/2 - \theta$ lies in $[0, \pi]$ — exactly the range of $\arccos$. So $\arccos(x) = \pi/2 - \theta$, which rearranges to $\arcsin(x) + \arccos(x) = \pi/2$, as required.

Differentiating a product involving arctan

- 23 Differentiate $y = x^2 \arctan(x)$ with respect to x .

By the product rule, $dy/dx = 2x \arctan(x) + x^2 \cdot \frac{d}{dx} \arctan(x)$. Since $\frac{d}{dx} \arctan(x) = 1/(1 + x^2)$, this gives $dy/dx = 2x \arctan(x) + x^2/(1 + x^2)$.

Evaluating arccos of a surd value

- 24 Find the exact value of $\arccos(\sqrt{3}/2)$, in radians.

$\arccos(\sqrt{3}/2) = \pi/6$, since $\cos(\pi/6) = \sqrt{3}/2$ and $\pi/6$ lies in the range $[0, \pi]$.

A composite that does not simplify to the inner angle

- 25 Find the exact value of $\arcsin(\sin(2\pi))$, and explain why it is not equal to 2π .

$\sin(2\pi) = 0$, so $\arcsin(\sin(2\pi)) = \arcsin(0) = 0$. This is not 2π because $\arcsin$'s range is restricted to $[-\pi/2, \pi/2]$, and 2π lies far outside that range — $\arcsin(\sin(\theta))$ only returns θ itself when θ is already inside $[-\pi/2, \pi/2]$.